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SireSoft Scan Tool

Project Documentation — company server par AI se receipt scanning

Version 1.0 | Roman Urdu | http://10.0.2.2:9000

Tayyar kardah
SireSoft Engineering

Tareekh
October 2026

Kis ke liye
Team, HR, naye developers

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1 Taaruf (Introduction)

SireSoft Scan Tool ek web application hai jo receipts aur bills khud parhti hai. User receipt ki photo ya PDF upload karta hai, aur system dukaan ka naam, tareekh, har item, tax aur total nikaal kar PostgreSQL database mein save kar deta hai.

Receipt parhne ka kaam ek AI vision model karta hai jo **company ke apne server** par chalta hai. Koi receipt internet par ya kisi bahar ki company ko nahin jaati. Poora system **Python** aur **Django** framework mein likha gaya hai.

> **Ek line mein:** `http://10.0.2.2:9000` kholen, login karein, receipt upload karein — chand second/minute mein uska data aapki History mein aa jata hai, aur 24 ghante baad khud delete ho jata hai.

Yeh document kis ke liye hai

Har us shakhs ke liye jo project istemal kare, samjhaye, chalaye ya aage barhaye — un logon ke liye bhi jinhon ne ise pehle kabhi nahin dekha. Mushkil alfaaz ke maani section 20 mein hain.

2 Maqsad (Goals and objectives)

Goal: receipts ka data haath se type karne ki jagah SireSoft staff ko ek tez, private aur apne server par chalne wali scanning service dena.

#	Objective	Kaise poora hota hai
O1	Receipts khud parhna	Local AI model (Ollama, qwen2.5vl:3b) har image ko structured data mein badalta hai
O2	Company ka data private rakhna	Model, database aur files sab server 10.0.2.2 par rehte hain
O3	Har user sirf apna data dekhe	Har page aur API receipts ko owner ke hisaab se filter karti hai; sirf admin sab dekhta hai
O4	Data zaroorat se zyada na rakhna	Har receipt upload ke 24 ghante baad delete, live countdown ke saath
O5	Galti theek karne ki sahoolat	User values edit, re-scan ya receipt delete kar sakta hai
O6	Admin ko poori nazar	History mein sab users; Django Administration mein asal database

3 Scope

Scope mein shamil

- Email se sign up, login, logout, password reset
- JPEG, PNG, WEBP, GIF aur PDF upload (ek dafa mein 5 files, har ek 10 MB tak)
- AI se vendor, address, tareekh, waqt, items, subtotal, tax, discount, doosre charges, grand total, payment method aur insights nikalna
- History, detail, edit, corrected copy, re-scan, delete
- Admin ko sab receipts; Django Administration
- Har receipt upload ke 24 ghante baad khud delete
- Company ke Linux server par 10.0.2.2:9000 par chalna

Abhi scope se bahar

- VPN ke baghair bahar se access
- Mobile app
- Excel ya accounting software mein export
- PDF ke saare pages (abhi sirf pehla page parha jata hai)
- Bohot halki likhai wali haath ki receipts
- Receipts ko lambe arse tak rakhna

4 Shuru karein: links aur login

Company VPN on karein, phir kisi bhi browser mein yeh address likhen:

`http://10.0.2.2:9000`

Aap kya chahte hain	Yeh link likhen
Login	`http://10.0.2.2:9000/auth/login/`
Naya account (Sign up)	`http://10.0.2.2:9000/auth/register/`
Receipt upload	`http://10.0.2.2:9000/`
Receipt History	`http://10.0.2.2:9000/history/`
Django Administration	`http://10.0.2.2:9000/admin/`
PostgreSQL mein saari receipts	`http://10.0.2.2:9000/admin/receipts/receipt/`
Saare users aur unki emails	`http://10.0.2.2:9000/admin/auth/user/`

Admin login

Kahan	Username / email	Password
App ka login page	`admin@siresoft.com`	Admin password, jo server ki `.env` file mein `ADMIN_PASSWORD` ke naam se hai. Project owner se poochen.
Django Administration	`admin` ya `admin@siresoft.com`	

> **VPN zaroori hai.** 10.0.2.2 company network ke andar ka private address hai. VPN ke baghair page nahin khulega.

5 Users, roles aur use cases

Role	Kya kar sakta hai	Kya dekhta hai
User	Sign up, upload, apni receipts dekhna, edit, re-scan, delete	Sirf apni receipts
Admin	User wala sab kuch, plus Django Administration	Har user ki receipts, "Uploaded By" column ke saath
System	AI model se receipts parhta hai; 24 ghante baad receipts delete karta hai	—

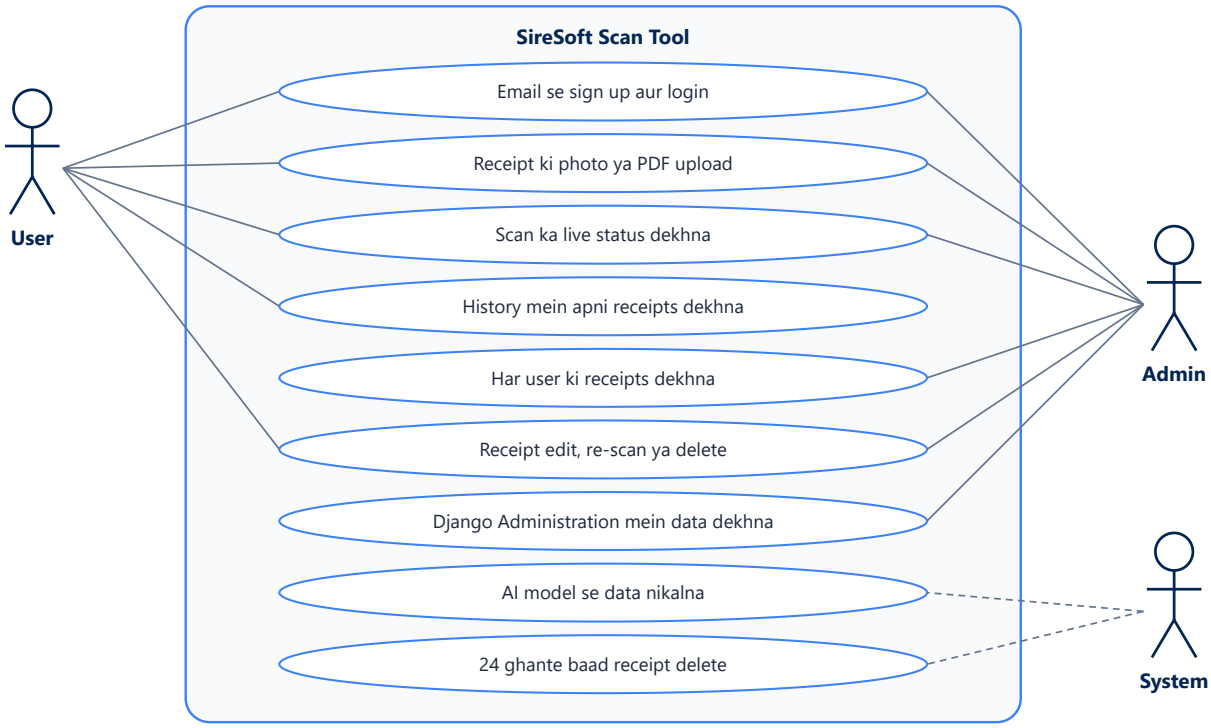


Figure 1 — Use case diagram

6 Requirements

Functional requirements (system kya karega)

ID	Requirement
FR-1	Visitor email aur password se account bana sakta hai aur usi email se login kar sakta hai.
FR-2	User ek dafa mein 5 tak files upload kar sakta hai (JPEG, PNG, WEBP, GIF, PDF; har ek 10 MB).
FR-3	System har receipt se vendor, tareekh, items, totals, payment method aur insights nikalta hai.
FR-4	Upload page har receipt ka status live dikhata hai: queued, reading, done ya failed.
FR-5	User sirf apni receipts dekhta hai; admin sab receipts dekhta hai.
FR-6	User nikli hui values edit kar sakta hai; system phir receipt ki corrected copy banata hai.
FR-7	User receipt re-scan ya delete kar sakta hai.
FR-8	Har user ki (admin samait) har receipt upload ke 24 ghante baad History, database aur disk se delete hoti hai.
FR-9	History mein har receipt ke saath live countdown ring baqi waqt dikhati hai.
FR-10	Admin Django Administration mein users aur receipts dekh sakta hai.
FR-11	User bhoola hua password email se reset kar sakta hai.

Non-functional requirements (system kaisa hoga)

ID	Khoobi	Requirement
NFR-1	Privacy	Koi receipt ya data company server se bahar nahin jata.
NFR-2	Security	Passwords hash ho kar save hote hain; har page ke liye login zaroori; koi doosre ki receipt nahin khol sakta.
NFR-3	Availability	Services server on hote hi khud chalti hain aur crash ke baad khud restart hoti hain (systemd).
NFR-4	Raftar	Upload foran wapas aata hai; parhne ka kaam background worker karta hai.
NFR-5	Data retention	Receipt apne 24 ghante poore hone ke taqreeban ek minute ke andar delete hoti hai.
NFR-6	Maintainability	63 automated tests; saari settings ek `.env` file mein.

Software aur hardware requirements

Hissa	Is project mein
Operating system (server)	Linux server **lsnet**, address 10.0.2.2
Programming language	Python 3
Web framework	Django (4.2 – 5.1) aur Django REST Framework
Database	PostgreSQL (driver psycopg2)
Web server	Gunicorn, systemd ke zariye; static files WhiteNoise se
AI model	Ollama par `qwen2.5vl:3b` (3.2 GB, CPU par bhi chalta hai)
Doosri libraries	PyMuPDF (PDF se image), Pillow (images), requests, python-dotenv
Client	Koi bhi naya browser (Chrome, Edge, Firefox), company VPN par
Hardware	Model ke liye kaafi RAM (taqreeban 4 GB khaali) aur uploads ke liye disk

7 System architecture

Project ke paanch hisse hain, sab ek hi server par. **Website** browser ko jawab deti hai, **worker** background mein AI se parhwane ka lamba kaam karta hai, **PostgreSQL** data rakhta hai, **AI model** images parhta hai, aur **media folder** upload ki hui files rakhta hai.

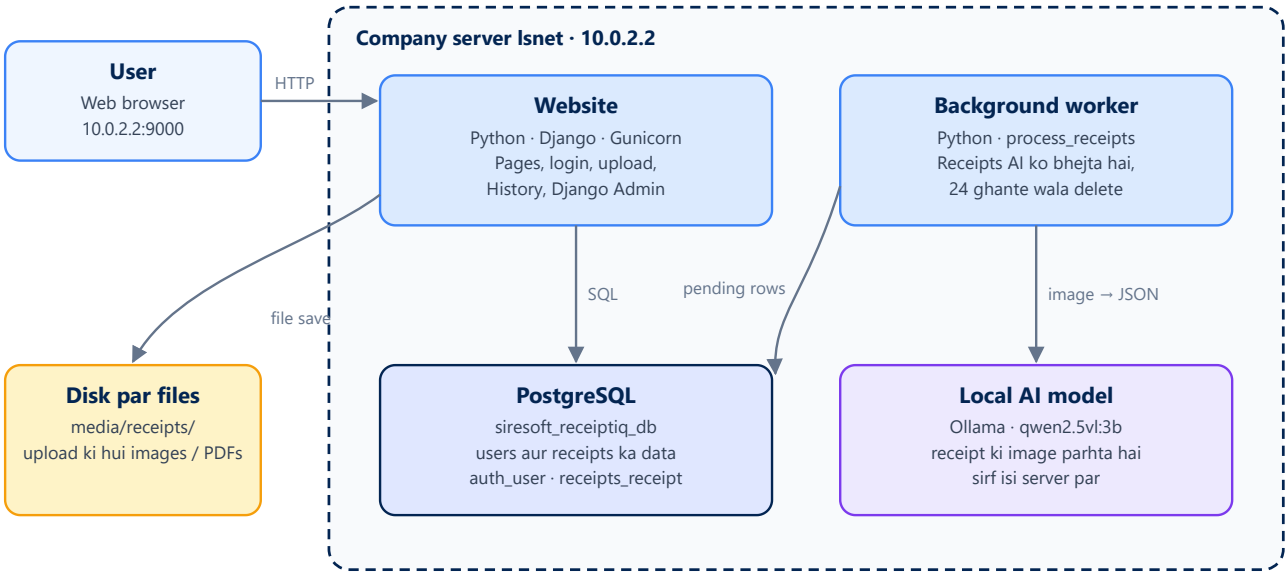


Figure 2 — SireSoft Scan Tool ka architecture

Front end aur back end

Layer	Technology	Ismein kya hai
Front end	HTML (Django templates), CSS, JavaScript	Login/sign up pages, drag-and-drop upload page aur live status, countdown rings wali History table, receipt detail aur edit pages
Back end	Python, Django, Django REST Framework	Accounts, receipts ka storage, permissions, upload API, AI extraction service, background worker, 24 ghante wala delete
Database	PostgreSQL	Users, receipts, nikala hua data, login sessions
AI	Ollama + qwen2.5vl:3b	Receipt ki image parh kar JSON deta hai

8 Project kaise kaam karta hai

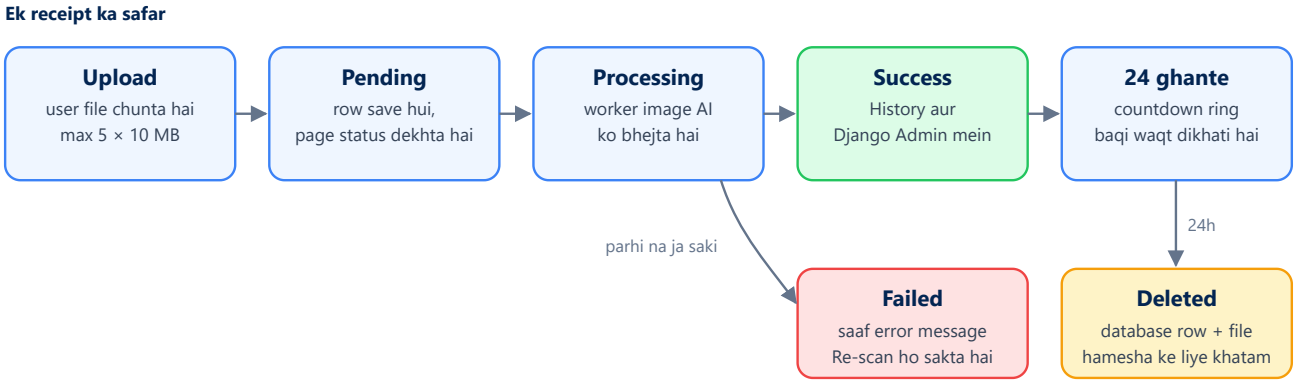


Figure 3 — Ek receipt ka lifecycle

1. **Upload.** User upload page par files daalta hai. Website type aur size check karti hai, file `media/receipts/` mein save karti hai, aur database mein *pending* status ki row banati hai. Page foran wapas aa jata hai.
2. **Queue.** Upload page har kuch second baad server se poochta hai ke har receipt ka kya haal hai (`/api/receipts/status/`).
3. **Processing.** Background worker sab se purani pending receipt uthata hai, use *processing* karta hai, PDF ho to image banata hai, bari image chhoti karta hai, aur AI model ko saaf hidayat ke saath bhejta hai.
4. **Nateeja.** Model JSON mein jawab deta hai. Worker use saaf karta hai (numbers, ghayab fields, totals) aur save karta hai. Status *success* ho jata hai, ya *failed* ek samajh aane wale message ke saath.
5. **Istemal.** Receipt History aur Django Administration mein aa jati hai. User use dekh, edit (corrected tasveer banti hai), re-scan ya delete kar sakta hai.
6. **Safai.** Upload ke theek 24 ghante baad receipt ki row aur files delete ho jati hain. History ki countdown ring batati hai kitna waqt baqi hai.

> **Background worker kyun?** Local AI model ko ek receipt parhne mein kuch second se kai minute lag sakte hain. Yeh kaam web request ke andar ho to page atak jaye; worker ki wajah se website tez rehti hai.

9 Data kahan store hota hai

Text data **PostgreSQL** mein jata hai; receipt ki files **server ki disk par ek folder** mein.

Kya	Theek kahan
Database server	PostgreSQL, 10.0.2.2 par
Database ka naam	`siresoft_receiptiq_db`
User accounts aur emails	Table `auth_user`
Receipts aur nikala hua data	Table `receipts_receipt`
Login sessions	Table `django_session`
Upload ki hui files	`/home/siresoft/siresoft-scan-tool/media/receipts/`
Corrected tasveerein	`/home/siresoft/siresoft-scan-tool/media/receipts/corrected/`
Connection settings	`DB_NAME`, `DB_USER`, `DB_PASSWORD`, `DB_HOST`, `DB_PORT` — `/home/siresoft/siresoft-scan-tool/.env` mein

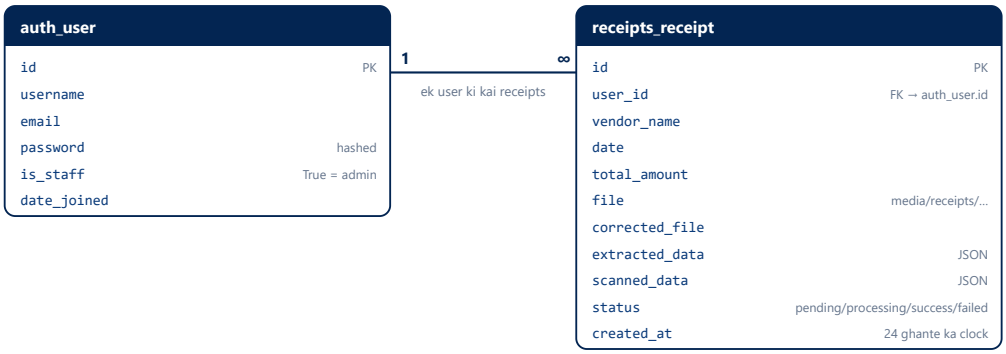


Figure 4 — Database design (entity relationship)

`extracted_data` ke andar kya hai

`scanned_data` AI ki pehli parhai ko waisa hi rakhta hai, taake edit ke baad har value pehle aur baad mein dikhayi ja sake.

```
{
  "establishment": { "name": "Y GENERATION SYSTEMS", "address": "...", "date": "25-Sep-2024",
"time": "..." },
  "items": [ { "name": "...", "quantity": 1, "unit_price": 3213.0, "total": 3213.0 } ],
  "bill_summary": { "subtotal": 0.0, "tax": 0.0, "discount": 0.0, "other_lines": [],
                    "grand_total": 3213.0, "payment_method": "cash" },
  "insights": [ "is receipt ke baare mein ek chhoti baat" ]
}
```

10 Django Administration

Django Administration ek tayyar admin website hai jo wahi PostgreSQL tables parhti hai. Data save hua ya nahin, yeh dekhne ka sab se aasaan tareeqa yahi hai.

1. VPN on karein aur `http://10.0.2.2:9000/admin/` kholen.
2. Username `admin` (ya `admin@siresoft.com`) aur admin password se login karein.
3. **Authentication and Authorization** ke neeche **Users** par click — har account, uski email aur sign up ki tareekh (table `auth_user`).
4. **Receipts** ke neeche **Receipts** par click — har receipt, vendor, total, status, owner aur upload ka waqt (table `receipts_receipt`).
5. Kisi bhi receipt par click karein — poora nikala hua data aur server par file ka theek path.
6. Daayein taraf ke filters (status, tareekh, user) aur search box (vendor) se list chhoti karein.

> **Sirf dekhne ke liye.** Django Administration mein receipts edit nahin hotin, kyunki worker ya user us tabdeeli ko mita dega. Receipt edit karni ho to app ke History page se karein.

11 PostgreSQL ke andar dekhna

Tables seedha dekhne ke liye server mein login karein aur database kholen:

```
# laptop par
ssh siresoft@10.0.2.2
# server par
sudo -u postgres psql siresoft_receiptiq_db
```

```
-- saare users aur unki emails
SELECT id, email, date_joined, is_staff FROM auth_user ORDER BY date_joined DESC;

-- aakhri 20 receipts, owner ke saath
SELECT r.id, u.email, r.vendor_name, r.total_amount, r.status, r.created_at, r.file
FROM receipts_receipt r JOIN auth_user u ON u.id = r.user_id
ORDER BY r.created_at DESC LIMIT 20;

-- har user ki kitni receipts
SELECT u.email, COUNT(r.id) FROM auth_user u
LEFT JOIN receipts_receipt r ON r.user_id = u.id GROUP BY u.email;

\q   -- psql se bahar
```

Save hui files dekhne ke liye: `ls -lh ~/siresoft-scan-tool/media/receipts/`

12 24 ghante baad khud delete

Har receipt ka apna 24 ghante ka clock hai jo upload hote hi shuru hota hai. Clock zero par pohanche to receipt History se, PostgreSQL se (yaani Django Administration se bhi) hat jati hai aur uski file server se delete ho jati hai.

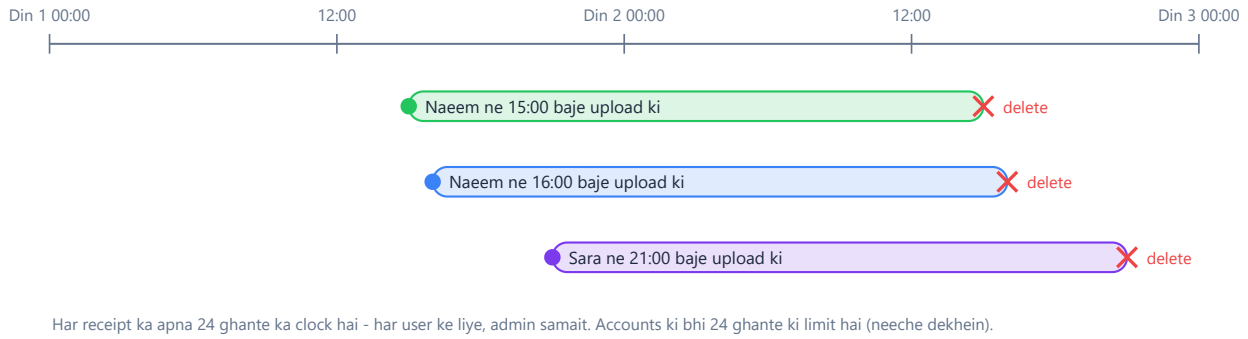


Figure 5 — Har receipt apne upload ke 24 ghante baad delete hoti hai

Usool	Tafseel
Clock kab shuru	Upload ke waqt (`created_at`)
Kab delete	24 ghante baad (`.env` mein `USER_DATA_TTL_HOURS=24`)
Kya delete hota hai	Database row, upload ki hui file, corrected tasveer
Kis par lagu	Har user ki har receipt, admin samait
Kya bachta hai	Sirf admin account (admin@siresoft.com)
Kitni dafa chalta hai	Har minute, worker aur website dono mein — cron ki zaroorat nahin
Haath se chalana	`./venv/bin/python manage.py purge_expired_receipts`

Countdown ring

History mein har receipt ke saath ek rangeen ring aur ghari jaisa waqt hai, maslan **23:54:12**, jo har second kam hota hai. Ring aam taur par **hari**, 3 ghante se kam par **peeli** aur 1 ghante se kam par **laal** hoti hai.

Sign up / sign in ki 24 ghante ki limit

Har user jo sign up karta hai, woh **sign up ke 24 ghante tak** hi sign in kar sakta hai. Har user ka clock alag hai aur sign up se shuru hota hai.

- 6:00 baje sign up kiya → agle din 6:00 baje tak sign in. 9:00 baje sign up kiya → agle din 9:00 baje tak.
- 24 ghante ke baad sahi password se bhi login nahin hota, aur jo user login ho woh agle click par logout ho jata hai.
- Account (username, email, password) aur us ki receipts PostgreSQL se delete ho jati hain. Login page par message aata hai: "Your 24-hour access has ended and the account was deleted. Please sign up again to continue."
- Wahi email dobara sign up kar sakti hai aur naye 24 ghante milte hain.
- **Admin par koi limit nahin.** Admin ke credentials server ki .env mein save rehte hain aur admin account kabhi delete nahin hota.

13 Project ka dhancha (structure)

Folder / file	Kaam
`apps/accounts/`	Sign up, login, password reset, admin account command (`ensure_admin`), 24 ghante wala delete (`retention.py`), Django Admin mein email se login
`apps/receipts/`	Receipt model, pages (upload, History, detail, edit), upload API, AI service (`services.py`), worker (`process_receipts`), corrected copy banana
`config/`	Django settings aur main URL list
`templates/`	HTML pages
`static/`	CSS, JavaScript, logos
`media/`	Upload ki hui receipt files (chalte waqt banti hain, Git mein nahin)
`deploy/`	Server ke start scripts aur systemd service files
`docs/`	Yeh documentation aur server commands ki sheet
`.env`	Raaz aur settings: database, admin account, AI model, port (Git mein kabhi nahin)
`requirements.txt`	Install hone wale Python packages

14 API endpoints

Upload page in endpoints se baat karta hai. Sab ke liye login zaroori hai, aur har user ko sirf apni receipts milti hain (admin ko sab).

Method	Address	Kaam
POST	`/api/upload/`	1–5 receipt files upload
GET	`/api/receipts/status/?ids=1,2`	Di gayi receipts ka live status
GET	`/api/receipts/`	Receipts ki list
GET	`/api/receipts/<id>/`	Ek receipt, poore data ke saath

15 Security aur privacy

- Login, sign up aur password reset ke ilawa har page ke liye login zaroori hai.
- Passwords Django one-way hash bana kar rakhta hai, kabhi saada text mein nahin.
- Receipts hamesha ek hi usool se dhoondi jati hain (`visible_to`): staff sab dekhta hai, baqi sirf apni. Kisi aur ki receipt ka address kholne par "not found" aata hai.
- Forms CSRF (cross-site request forgery) se mehfooz hain. Purana form (back button, resubmission) error ki jagah taaza page par le jata hai.
- Raaz (database password, admin password, secret key) sirf server ki `.env` file mein hain, jo Git mein nahin jaati.
- AI model local chalta hai; koi receipt bahar ki service ko nahin bheji jati.
- Receipts aur users ke accounts sirf 24 ghante rakhe jate hain, is liye kam personal data store hota hai.

16 Testing

Project mein **63 automated tests** hain jo ek aarzi test database par chalte hain. Project folder se chalayen:

```
python manage.py test apps
```

Test group	Kya check karta hai
LoginViewTests, RegisterTests	Email ya username se login, sign up ke usool, ek email do dafa nahin
EnsureAdminTests	Admin account `.env` se banta aur theek hota hai
RetentionTests	Har receipt apne upload ke 24 ghante baad delete; files bhi hat-ti hain; admin ki receipts bhi; account signup ke 24 ghante baad khatam; admin Django Admin mein email se login kar sakta hai
ReceiptVisibilityTests	User doosron ki receipts na dekh sakta hai na khol sakta hai; admin sab dekhta hai
ReceiptFileTests, ReceiptPreviewTests	Files mehfooz tareeqe se dikhayi jati hain; PDF ka tasveeri preview
ReceiptEditTests, CorrectedCopyAndDeleteTests	Values edit, corrected copy, receipt aur uski files delete
ExtractionDetailsTests, RescanTests	AI ke jawab ki safai aur re-scan

Python 3.14 par login page ke teen tests Django ke test client ki ek maloom kharabi ki wajah se error dete hain; chalti hui application par iska koi asar nahin.

17 Deployment aur server

Cheez	Value
Server	lsnet, `10.0.2.2`, user `siresoft`
Address	`http://10.0.2.2:9000`
Project folder	`/home/siresoft/siresoft-scan-tool`
Services	`siresoft-scan-tool` (website), `siresoft-scan-tool-worker` (AI worker)
Code repository	GitHub: Engr-Mahnoor-code/SireSoft-Scan-Tool (private)

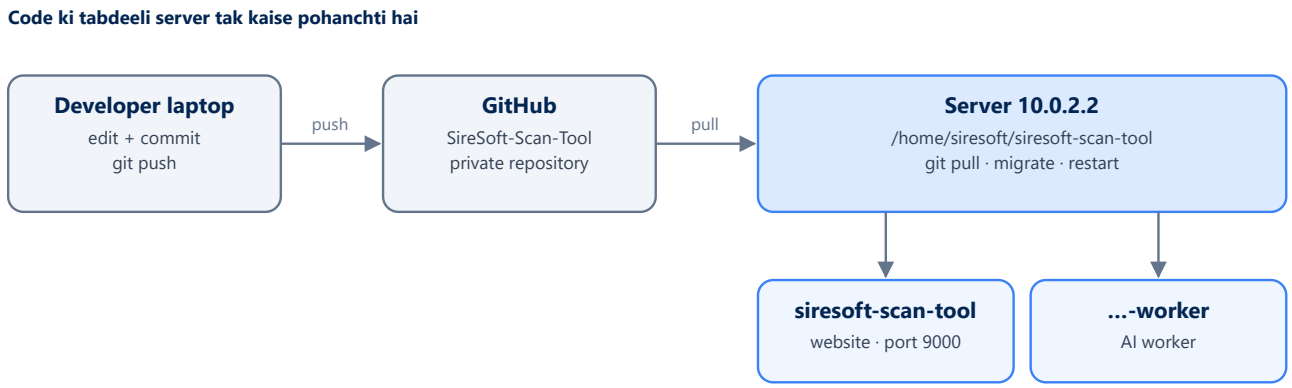


Figure 6 — Code ki tabdeeli se chalte server tak

Rozana ke commands (server par)

```
# chal raha hai? do dafa "active" aana chahiye
systemctl is-active siresoft-scan-tool siresoft-scan-tool-worker

# restart
sudo systemctl restart siresoft-scan-tool siresoft-scan-tool-worker

# logs
journalctl -u siresoft-scan-tool -n 50 --no-pager
journalctl -u siresoft-scan-tool-worker -n 50 --no-pager
```

Code ki tabdeeli server par lagana

```
# 1. laptop par (prompt: DC@Mahnoor MINGW64)
git add -A
git commit -m "tabdeeli ka bayan"
git push origin main

# 2. server par (prompt: [siresoft@lsnet ...]$)
ssh siresoft@10.0.2.2
cd ~/siresoft-scan-tool
git pull
./venv/bin/python manage.py migrate
./venv/bin/python manage.py collectstatic --noinput
sudo systemctl restart siresoft-scan-tool siresoft-scan-tool-worker
```

18 Masle aur unka hal

Masla	Mumkina wajah	Hal
Page nahin khulta	VPN band hai	VPN on karein, phir http://10.0.2.2:9000 kholen
10.0.2.2:9000 par koi aur app khulti hai	Port 9000 kisi aur service ne le rakha hai	`sudo ss -tlnp | grep :9000`, woh service band karein, Scan Tool restart karein
Service "activating" dikhati hai	Website start nahin ho pa rahi	Website ka log parhen
"Server Error (500)"	App ke andar error	Website log ki aakhri lines wajah batati hain
Receipt "Queued" par atki hai	Worker ya Ollama nahin chal raha	Worker ka log aur `ollama list` check karein
Receipt "Failed"	Tasveer saaf nahin	Behtar photo upload karein ya Re-scan
Nayi styling nazar nahin aati	Static files collect nahin huin	`collectstatic`, restart, phir Ctrl+Shift+R
"No such file or directory"	Command server ki jagah laptop par chala	Pehle `ssh siresoft@10.0.2.2`
Admin login nahin hota	Database ka password .env se mukhtalif	`./venv/bin/python manage.py ensure_admin`

19 Aage ke goals

Idea	Faida
Mehfooz public address (HTTPS domain, maslan receipts.siresoft.com)	VPN ke baghair istemal
Excel / CSV export	Ek click mein data accounts ko
Multi-page PDF ke saare pages parhna	Poori invoice, sirf pehla page nahin
Vendor aur mahine ke hisaab se totals ka dashboard	Management ke liye kharche ki tasveer
User ya team ke hisaab se retention ka waqt	Zaroorat ho to kuch receipts zyada der rakhna
GPU par bara model	Halki receipts ki tez aur behtar parhai
Mobile camera se upload	Phone se seedha receipt scan

20 Alfaaz ke maani (Glossary)

Lafz	Matlab
Python	Woh programming language jis mein poora project likha gaya hai
Django	Websites banane ka Python framework; login, database aur admin site deta hai
Django Administration	/admin/ par Django ki apni admin website, save hua data dekhne ke liye
PostgreSQL	Database program jo users aur receipts tables mein rakhta hai
Table / row / column	Table Excel sheet jaisi; har receipt ek row; har field ek column
Ollama	Program jo AI models hamare apne server par chalata hai
qwen2.5vl:3b	Vision AI model jo receipt ki tasveer parhta hai
Gunicorn	Program jo Django website ko port 9000 par chalata hai
systemd service	Linux ki khoobi jo programs ko server on hote hi chalati aur rukne par dobara chalati hai
Worker	Background program jo receipts ek ek kar ke process karta hai
IP address / port	10.0.2.2 server ka address; 9000 woh "darwaza" jis par website sunti hai
VPN	Company network mein mehfooz connection, 10.0.2.2 tak pohanchne ke liye zaroori
.env file	Server par private settings ki file (passwords, database ka naam)
Git / GitHub	Code ka version control / woh website jahan repository rakhi hai
JSON	Structured data ka text format, receipt ka nikala hua data isi mein